

EEG-based Adaptive Visual Alerts to Mitigate UAS Operator Mental Fatigue: A Technical Proof of Concept

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Abstract

Uncrewed Aerial Systems (UAS) mishaps are partially attributable to mental fatigue during long endurance missions. Mental fatigue decreases cognitive flexibility resulting in a decreased performance when switching between sub-tasks. Solutions include adaptive interfaces. We propose a technical proof of concept that launches visual alerts to mitigate operator mental fatigue by estimating future performance based on brain activity (EEG). The proposed solution uses flying and detection tasks, and includes an online classification pipeline. The method is designed to work in an intra-user and inter-session manner, i.e. the solution is tuned to a user and works without calibration in the following sessions. The proof of concept was validated on three participants and compared with a previous campaign with randomly launched alerts. Very promising results were found (negative correlation between visual alerts activation and detection misses). This opens the way for the use of physiology-based adaptive interfaces to enhance user performance and safety.

CCS Concepts

• Human-centered computing → Laboratory experiments.

Keywords

UAS, drones, adaptive interfaces, physiological computing, EEG, mental fatigue, alerts

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1 Introduction

In our daily lives, multitasking is almost a prerequisite for a great variety of jobs. Sustained attention requirements, on the other hand, are very specific to safety critical operations such as driving, but

also long-endurance operations of Uncrewed Aerial Systems (UAS)¹. Yet, prolonged task performance is known to cause mental fatigue and performance degradation with increased reaction times and decreased accuracy [19, 52]. Operator fatigue has been reported as a major issue in UAS operations [2, 6], and UAS mishaps are at least partially attributable to mental fatigue during long-endurance missions. This mental fatigue induces a decrease in cognitive flexibility that can result in a decreased performance when switching from one subtask to another [38, 47, 53].

A simple solution to address the issue of mental fatigue would be more frequent rotations between teams operating a single UAS. The problem is that this would require many more operators to operate one UAS, negating one of the great advantages of UAS (i.e., cost). Another problem is that hand-off procedures between teams are a large risk factor during operations [3, 51].

When shift work organization prevents taking breaks, solutions to mitigate this phenomenon while maintaining mission time could include monitoring operators' mental state and taking action when performance degradation is predictable from their current state. Operator monitoring based on physiological measures is a growing field of research, and has been dubbed 'physiological computing' [1, 15] and 'passive brain-computer interfaces' [14] when using respectively peripheral (e.g., cardiac, ocular, electrodermal) and central nervous system measures which are then objectively analyzed using machine learning tools [12, 32]. These monitoring tools can then be used to adapt the interaction between the user and the system.

A solution to try to mitigate a degraded operator mental state to avoid degraded performance is the use of adaptive user interfaces. Adaptive interfaces are user interfaces that automatically adapt to a user's capacities or knowledge of a system or context of use [5] (contrary to adaptable user interfaces, where the adaptation is done by the user [13]). Previous review work suggests the use of adaptive interfaces in neuro-adaptive technology to tackle degraded user states [11]. In the context of aeronautics, a simple visual alert proved successful in mitigating attentional tunneling of air traffic controllers in situations of high mental workload [45].

In a prior recent study we demonstrated that mimicking an adaptive interface with intermittent visual alerts on a UAS piloting interface significantly increased operator performance during a 2-hour mission compared to a control condition without any visual alerts [20]. In this study, alerts warned users about a switch in the

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¹UAS are aircraft that do not house a pilot within the airframe and are operated remotely using various levels of automated functions [16]. For the sake of inclusivity the general naming convention is changing from Unmanned to Uncrewed [17].

sub-task, which allowed them to increase their accuracy while response time, on the other hand, increased due to the duration of the alert. Although the usefulness of these alerts has been demonstrated in general for this type of task, it remains to ascertain whether a real adaptive interface based on an online assessment of operator fatigue and performance could improve cognitive flexibility during long-endurance UAS missions.

Hence, this article presents a technical proof of concept of the implementation and validation of an adaptive interface based on cerebral activity. To the best of our knowledge, no such interface has yet been proposed in the literature. This interface is designed to launch visual alerts during a UAS piloting task to warn users about a change in subtask. Input measures are cerebral markers extracted from electroencephalography (EEG). These markers are classified with a state-of-the-art estimation pipeline to predict user performance. The whole system is closed-loop, meaning that the system adapts based on the users' state without any outside input. We validated its effectiveness by a user study, the results of which are compared with the ones of the previous study [20] that comprised a 'no interface adaptation' and a 'sham adaptation' (i.e. mimicking a real adaptation) conditions.

The structure of this article is as follows: the proposed technical solution is first described below, followed by the validation of this proof of concept by a user study, and then the results are discussed.

2 Proposed technical solution

While human-drone interaction is a growing topic in the HCI community [7, 18] and professional drone piloting has been studied [23, 34, 42], to the best of our knowledge the literature does not yet report an adaptive interface based on cerebral activity to mitigate mental fatigue and cognitive flexibility issues during UAS operations. Hence, this study is a proposition to fill this gap. It proposes a technical proof of concept of an adaptive interface that launches visual alerts to mitigate UAS operator mental fatigue, based on an estimation of operator future performance from cerebral activity (measured by EEG). The proposed technical solution uses an open access UAS task with flying and detection sub-tasks [22], and includes an online classification pipeline that performs operator performance estimation based on EEG covariance matrices and a TSMnet classifier (see details in dedicated section 2.3). This method is designed to work in an intra-user and inter-session manner: this means that the solution is tuned to a user and then works without calibration in the following sessions of the same user.

2.1 UAS task

The open access UASOS (Unmanned Aerial Systems Operator Simulator) task from [22] combines some of the fundamental aspects of UAS operations with a task-switching protocol to allow the investigation of cognitive flexibility (see Fig. 1). The UASOS task requires users to alternate between sub-tasks on a trial-based system. On average, every 7 seconds (with a ± 1000 ms jitter) written instructions appear on a widget in the middle of the user's visual field to indicate the current sub-task (Fig. 2 left). The two main sub-tasks that users work on (Navigation and Search) have two modes each. The Navigation task requires the user to navigate the UAV either using headings (Heading mode, HDG) or waypoints

(Waypoint mode, WPY) (right screen in Fig. 1 and Fig. 3 below). The Search task requires the user to search for either humans or vehicles in a search and rescue scenario (left screen in Fig. 1 and Fig. 3 top; images from the open NORB—NYU Object Recognition Benchmark—database from [33]). When authors refer to repetition trials, it means the absence of instruction change for the trial at hand (as compared to the previous one); when they refer to internal switches they consider a change in instruction within one sub-task, while external switches refer to a change between sub-tasks.

2.2 Adaptive interface implementation

In the proposed solution, the UASOS interface has been adapted by inserting visual alerts to warn users of a forthcoming task change as suggested by prior work [45]. Two changes were triggered when visual alerts became active. First, the visual alerts widget was turned on to inform users (Fig. 2 right part). Second, the activation of the visual alerts themselves depended on the upcoming trial. During repetition trials, when the previous sub-task was the same as the upcoming one, nothing happened. 700 ms before switch trials (when changing from one sub-task to another), a transparent amber  (R: 254, G:80, B:0; Opacity: 50%) flash would simultaneously appear on both computer screens (for 50 ms; see Fig. 4). Activation of visual alerts was user-dependent: the activation schedule for visual alerts was based on users' estimated performance (linked to mental fatigue), for which their real-time EEG activity was used, as explained in the next section.

2.3 Online EEG data processing

We performed the following data processing to launch visual alerts depending on performance:

- **Preprocessing:** The data from 64 EEG electrodes are split into 10-second epochs and average referenced with zero padded channel. The data are then bandpass filtered between 1 and 40 Hz using the `mne.filter()` function. Then they undergo a downsampling at 125 Hz (factor 4).
- **Estimation:** For the classification stage, the covariance matrices—i.e., state-of-the-art EEG features for passive brain-computer interface applications used with Riemannian geometry-based estimation methods [44]—are directly fed to a TSMnet classifier [30]. The training of this classifier is performed using the data from a given user on a given session (e.g., session 1) and the testing set consists of any subsequent session (e.g., session 2). No calibration is required for applying this classifier to the testing set as this method has been shown to handle EEG non-stationarities and provides therefore an adequate enough transfer learning method for our proof-of-concept [30].
- **Labels:** As for the labels used for classification, the performance to the task is modeled using the misses to the search sub-task. The performance labels are normalized misses that are computed using a moving average. Only the best 40% and the worst 40% are calculated for the training set (i.e., same as in the previous study by [20]).

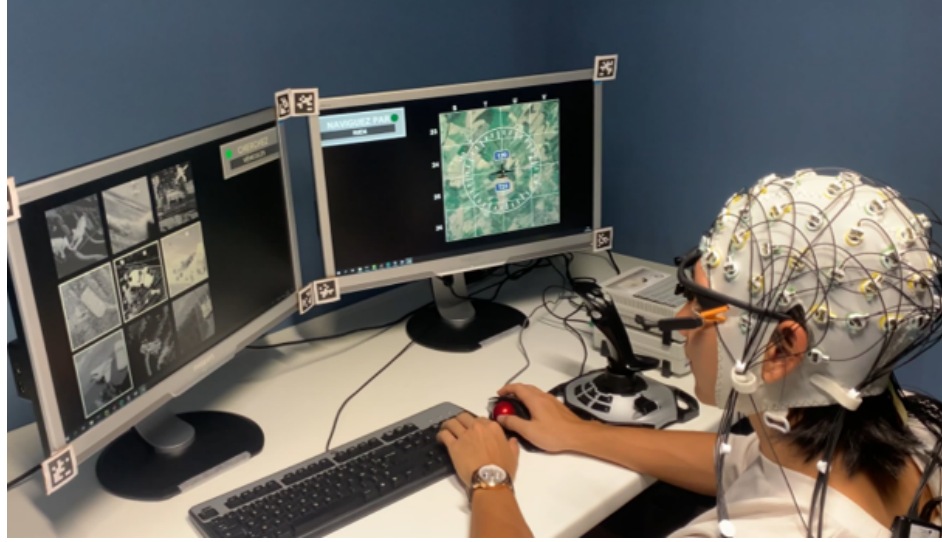


Figure 1: Participant equipped with the EEG device and performing the UASOS task (NB: eye-tracker worn for checks but not used for online operator state estimation).

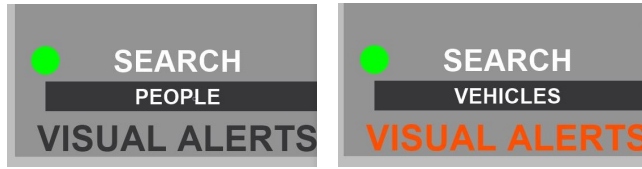


Figure 2: Flight director widget for the Search task. Left: visual alerts are deactivated; Right: visual alerts are activated, and the user is informed of this via the amber hue of the Visual Alerts widget.

3 Validation

The proposed adaptive interface based on EEG was validated by running a user study with three participants. This dataset will be called the ‘Online’ condition of the validation study.

The results obtained with it are here compared to that of our prior recent study run on 30 participants [20]. In this prior study we demonstrated that mimicking an adaptive interface with intermittent randomly launched visual alerts significantly increased operator performance during a 2-hour mission compared to a control condition without any visual alerts [20]. In this study, alerts warned users about a switch in the sub-task, which allowed them to increase their accuracy while response time, on the other hand, increased due to the duration of the alert. This previous campaign will be called the ‘Offline’ campaign. The control condition of this previous study will be called ‘Control’ (with no alert), and the condition with randomly launched alerts that mimic an adaptive system will be called ‘VA’ (Visual Alerts). These two datasets (Online and Offline) are detailed below in their respective subsection (i.e., Dataset and Comparison dataset).

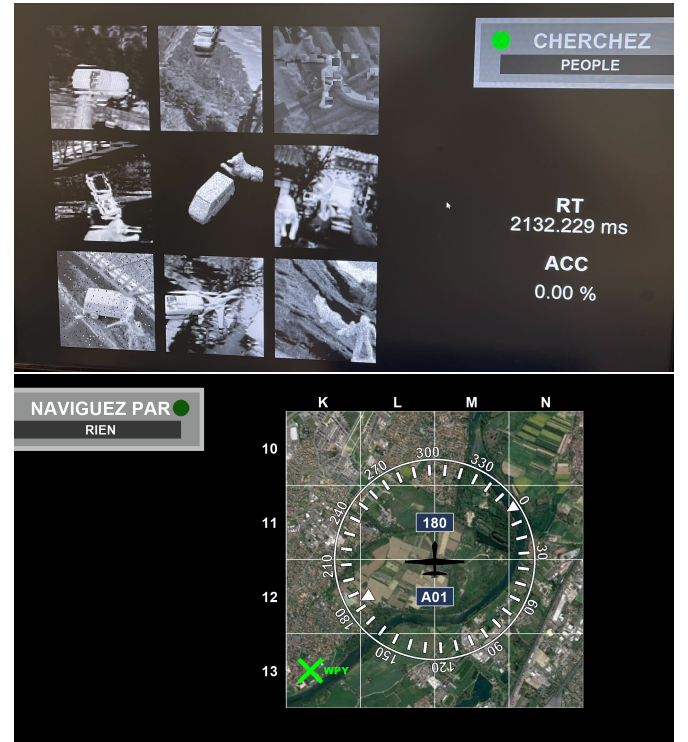


Figure 3: Example of the screens (French version) for the search task (top) and the navigation task (below).

3.1 Dataset

3.1.1 Ethical validation. The ethical committee of the University of Toulouse approved the experimental protocol (file number 2022-501).

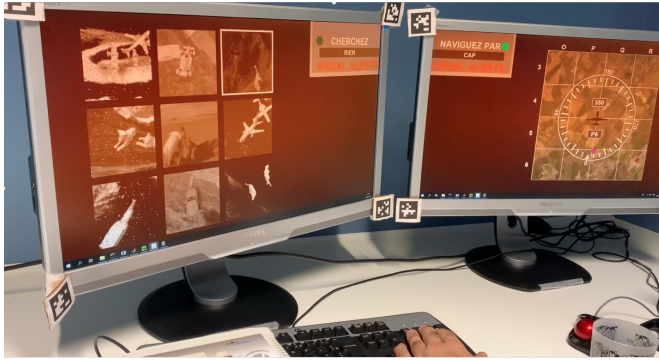


Figure 4: Visual alert (amber flash displayed on the UASOS interface when operator state is classified as degraded and likely to yield poor performance. The alert indicates that a change in instruction is about to occur.

3.1.2 Participants. Three participants (1 female, 2 male) agreed to take part in the study. The participants were all right-handed PhD students, on average 28.33 years old.

3.1.3 Experimental protocol. Using an active AG-AgCl electrode system with an ActiCHamp amplifier (Brain Products, GmbH), EEG data were recorded from 64 electrodes. The international 10–20 system was used for electrode placement [26]. Data were recorded at 500 Hz and impedances were kept below 50 k Ω . Data were streamed and synchronized using the Lab-StreamingLayer (LSL) [31].

Participants signed the informed consent form and were equipped with the EEG device. Next, they completed a battery of questionnaires including demographics, handedness (Edinburgh Handedness questionnaire - EDI) and mental fatigue (Karolinska Sleepiness Scale - KSS; Samn-Perelli Scale - SPS) [28, 46, 49]. Next, they completed a training phase of 16 minutes before a five-minute resting state was recorded (30-second intervals of eyes open and closed). They then started the 120-minute main phase of the experiment. After completing the main phase, participants completed another battery of KSS and SPS questionnaires, as well as the RSME questionnaire for mental effort.

3.2 Comparison dataset

As detailed at the beginning of the Validation section, in order to appropriately ascertain the operation and effectiveness of the proposed technical solution with EEG-based visual alerts' activation, the results obtained with the Online campaign were compared to that obtained by [20] with their Offline campaign (30 participants, 2-hour session of UASOS task). In this previous campaign, the use of visual alerts in the UASOS interface was tested by comparing a control condition with no alert to a condition with only a sham adaptive interface, that is to say, with visual alerts that were randomly launched without real user state estimation. Hence, in this new study, a real adaptive interface is tested and both behavioral and subjective results were directly compared for the three conditions of the two campaigns: Online, Control and VA (resp. real, no, and sham adaptive interfaces).

Participants of the Online campaign were recruited from the participants of the Offline campaign, VA condition. Hence, the estimation pipeline was trained on the first session (Offline VA; all data from the 2-hour session) and tested on their second session in an online manner (Online), with an intra-user inter-session estimation (a.k.a within-participant cross-session estimation). This enabled us to demonstrate the practicality of the technical solution with no need for calibration for any other session when training of the pipeline is done on session 1.

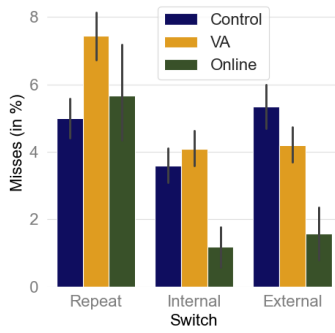
3.3 Results

This section details the behavioral and subjective results that validate the proposed EEG-based adaptive interface. As data from only three participants were collected at this stage for the Online campaign, no statistical analyzes could be performed at the group level. However, a correlation analysis was possible to perform between the activation of visual alerts during the test phase (session 2) and the performance to the detection sub-task, as illustrated below. Apart from this correlation analysis performed solely on data from the Online campaign, all other results are computed with respect to the Offline campaign (Control and VA conditions). As a reminder, repeat trials correspond to trials in which no change in instruction occurs, internal switch corresponds to a change in instructions for the same sub-task, while external switch corresponds to a change in sub-task.

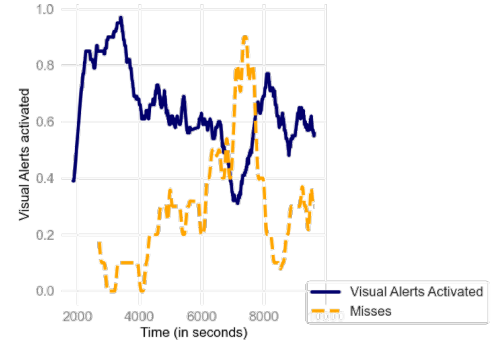
3.3.1 Behavioral results. The performance of the participants in the Online campaign showed a strong increase for the internal and external switches as compared to the Offline campaign (control and random alerts), as illustrated by Figure 5a. Yet, this phenomenon can be explained in no small part by the fact that for the Online campaign, participants came for two sessions—session 1 served as the training set and session 2 as the testing set—while in the previous Offline campaign participants had only one session. This increase in performance in the Online campaign could therefore reveal a training effect.

However, very interestingly, when computing the Pearson correlation of the activation of the visual alerts during the Online campaign and the performance to the detection sub-task (misses), a negative correlation of -0.314 in average across the three subjects was found. The best subject had even a -0.74 correlation with a visible drop in performance when alerts were deactivated, and increase in performance when they were activated (Figure 5b). These results can be compared to the correlation obtained in the Visual Alerts (VA) condition of the Offline campaign, in which alerts were randomly activated, with only a -0.032 correlation with performance. These correlation results in themselves validate the effectiveness of the proof of concept.

3.3.2 Subjective results. Subjective results obtained from the Karolinska Sleepiness Scale—KSS (Fig. 6a)—and the Samn-Perelli Scale—SPS (Fig. 6b)—questionnaires show that no major modification of reported fatigue can be seen between conditions 'Online' and 'VA' in the end of the experiment, as for both of them reported fatigue appears lower than the control condition with no visual alert. Interestingly, the reported mental effort was lower for the 'Online' condition compared to the control and 'VA' conditions (Fig. 6c).



(a) Percentage of misses to the detection sub-task at the group level for both the technical proof-of-concept campaign (Online) and the previous campaign that used a control condition without alerts (Control), and a condition with randomly launched alerts (VA).



(b) Percentage of time of activation of visual alerts and misses to the detection task during the test phase (i.e. session 2) for the best participant (Pearson correlation of -0.74). Note that the first miss appeared later than the first visual alert (hence the graph for misses starting later).

Figure 5: Validation user study – Behavioral results.

This could reflect an ease of use of the adaptive interface, but might also reflect the fact that participants for the ‘Online’ condition had already encountered the interface in a previous session.

4 Discussion

The main goal for this study was to propose a proof-of-concept of an adaptive interface for UAS operations that would mitigate user mental fatigue and its associated performance decrease by launching visual alerts based on an estimation of user state thanks to cerebral measures. To the best of our knowledge, this technical solution had never been reported in the literature.

Its operation and effectiveness are demonstrated here by conducting a validation study and comparing it to a previous study that only mimicked an adaptive interface with the same UAS operation task [20]. Behavioral results revealed an increase in user performance when visual alerts were activated and a decrease when they were deactivated with a significant negative correlation. In comparison, no correlation was observed in the previously published study in which alerts were launched at random. Very interestingly, one can see that mental fatigue was still present in all conditions—as underlined by the subjective results—, which means that the proposed solution that predicts performance using EEG data successfully mitigated the impact of mental fatigue on user performance.

As revealed by both the Online and the Offline studies, the mere presence of alerts did increase general performance. This phenomenon could be due to a salience effect that would need further investigation [40], without forgetting the alert fatigue effect that might occur if the frequency of alert activations is too high [27].

This proof-of-concept presents preliminary results and should be followed by a larger campaign with more participants and several sessions each, in order to tackle the possible learning effect to consolidate the results presented in this paper. In addition, using other modalities for the alerts should be investigated to tackle other mental states that could deteriorate user performance, such as inattention blindness or deafness [11].

4.1 Physiological computing and passive brain-computer interfaces and the HCI community

In this study, we focused on an adaptive system using EEG data for one specific use case, namely UAS operations and user performance under mental fatigue issues. It would be of particular relevance to investigate whether the proposed passive brain-computer interface solution could be effective for other human-robot interaction (HRI) tasks, for instance, for various degrees of autonomy in teleoperation applications, up to interaction with autonomous robots [43].

Moreover, it could be investigated for other types of mental fatigue. Mental fatigue elicited by prolonged task performance has been extensively studied with respect to user interaction with critical systems [19]. However, this type of mental fatigue may be easier to mitigate through adaptive interfaces than mental fatigue and related cognitive dysfunctions elicited by clinical states such as post-infectious syndromes (e.g., long covid / post-acute COVID-19, or myalgic encephalomyelitis/chronic fatigue syndrome; [4, 41]). Yet, despite a growing interest these clinical applications are still in dire need of research work in all domains including HCI, and would yield significant life quality improvement [36, 37]. User/patient monitoring solutions such as pacing tools could for instance help patients manage their activity level to reduce the occurrence of post-exertional malaise, as recently proposed by [24].

Furthermore, this work on an adaptive system that uses user’s physiological data to enhance interaction contributes to a growing interest in the research community for studies of this vein and should be of interest to the broader HCI community. Indeed, systems that make use of cerebral data for adaptive solutions have recently been designed to detect other relevant user mental states—such as mental workload or engagement—, possibly with other devices than EEG—for instance functional near infra-red spectroscopy (fNIRS)—and for other applications—such as optimizing multitasking, personalizing cultural heritage experiences virtual reality or player immersion [8, 9, 29, 35, 39, 50].

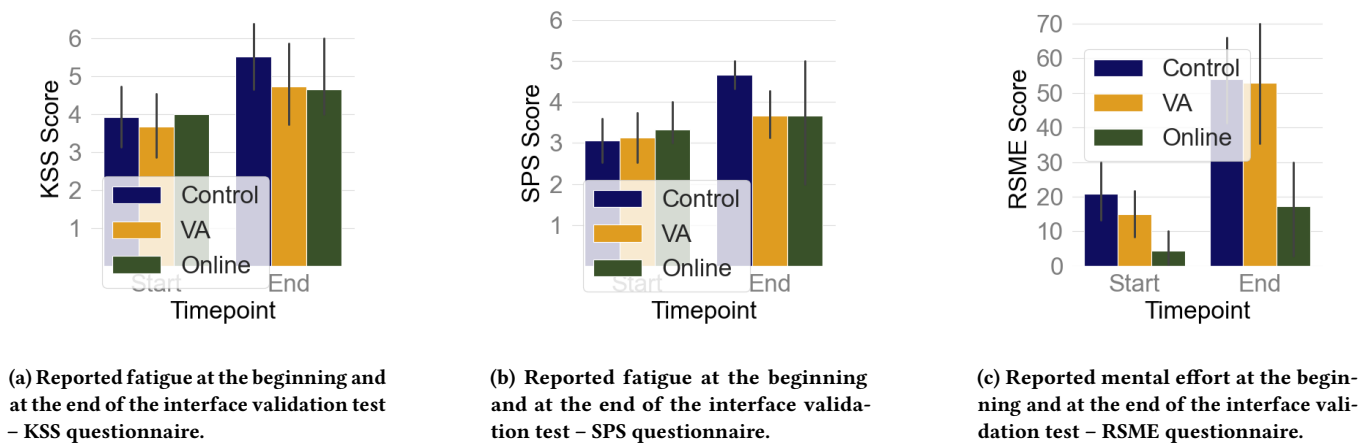


Figure 6: Validation user study – Subjective results.

In addition to code sharing, physiological data sharing is also vital to foster research in an open science perspective, such as recently initiated for both EEG and fNIRS data [21, 25]. Hence, to meet the needs and interests of the HCI community, and more specifically the growing communities of PhysioCHI [10] and NeuroCHI [48], efforts must be made to consolidate this research and extend it beyond its original field of application.

5 Conclusion

This study illustrates a proof-of-concept of how the use of passive brain-computer interfaces and physiological computing can be combined with adaptive interfaces to mitigate operators' mental fatigue and enhance their performance. This constitutes a stepping stone to safer and more effective UAS operations, and a concrete example of the usefulness of passive brain-computer interfaces and adaptive interfaces in the HRI and aerospace domains. As such, it contributes to the growing communities of PhysioCHI [10] and NeuroCHI [48], as well as Human-Drone Interaction [7].

Acknowledgments

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